

PROOFLEARN

Standard Operating Procedure (SOP)

"Don't just get the answer. Prove you learned it."

Document Type	Project Standard Operating Procedure
Project	PROOFLEARN – AI-Powered Learning Verification SaaS
Version	1.0
Purpose	Define the end-to-end operating process for learning, practice, proof, transfer, evidence and history.
Primary Principle	AI should help students learn, not replace their ability to think.

Prepared from the uploaded PROOFLEARN project documentation.

1. Purpose

This SOP defines the standard operating flow for PROOFLEARN. The platform is designed to move a learner from AI-assisted learning toward independently demonstrated understanding. The process ensures that a correct answer alone is not treated as proof of mastery.

2. Problem Statement

Generative AI can provide correct answers instantly, but a correct answer does not necessarily demonstrate independent understanding. PROOFLEARN addresses three major risks: illusion of competence, fragile knowledge when the context changes, and evaluation blindspots about whether the learner or AI actually solved the problem.

Core principle: Correct Answer $\neq$ Independent Understanding

3. Scope

- Student learning through a Socratic AI learning room.
- Scaffolded formative practice and knowledge checks.
- Server-enforced PROOF MODE for independent assessment.
- Independent challenge and novel transfer challenge.
- Calculation of the Learning Evidence Index (LEI).
- Recording of verified learning attempts in the student-owned history ledger.
- Secure authentication, data isolation, rate limiting and secret protection.

4. Roles and Responsibilities

Role	Responsibility
Student / Learner	Learn concepts, complete practice, enter Proof Mode, solve independently, complete transfer tasks, and review
AI Tutor	Provide Socratic guidance while AI access is permitted; support learning without replacing independent thinking
Proof Guard	Enforce server-side AI lockout during an active proof session.
LEI Engine	Synthesize recall, explanation, application, transfer and independence signals into a bounded evidence score
System / Backend	Authenticate users, validate ownership, enforce security controls, process learning data and persist evidence.

5. Standard Operating Workflow

Step 1 – LEARN

The student studies a selected concept in the Socratic AI Learning Room. AI guidance is allowed at this stage.

Step 2 – PRACTICE

The student completes formative knowledge checks using scaffolded practice questions and guidance.

Step 3 – PROOF MODE ACTIVATION

The system activates a server-side AI lockout. During an active proof session, AI learning access is blocked at the API layer.

Step 4 – PROVE

The student solves an independent challenge without AI assistance. The response becomes evidence of independent problem application.

Step 5 – TRANSFER

The student applies the same underlying concept in an unfamiliar or novel context to test conceptual transfer.

Step 6 – EVIDENCE

The system synthesizes multiple signals into the Learning Evidence Index (LEI), bounded from 0 to 100.

Step 7 – HISTORY

The verified attempt and evidence breakdown are recorded in the private, student-owned Learning History ledger.

6. Learning Evidence Index (LEI)

LEI is a prototype product metric representing demonstrated engagement and performance. It is not an IQ score, psychometric assessment, accredited academic credential, or replacement for formal evaluation.

Signal Dimension	Weight	Evaluated Capability
Concept Recall	15%	Immediate terminology and core syntax recall
Socratic Explanation	20%	Ability to articulate reasoning and explain why
Problem Application	20%	Accurate implementation in an independent challenge
Novel Transfer	25%	Application of a concept in an unfamiliar scenario
AI Independence	20%	Completing challenges without AI reliance

The documented model is: $LEI = \max(0, \min(100, \sum(w_i \times s_i) - \text{Penalty}))$.

7. Security and Trust Procedure

- Use verified Supabase JWT identity rather than client-supplied identity fields.
- Enforce Row Level Security (RLS) for tenant isolation.
- Validate student ownership for sessions, answer submissions and evidence retrieval.
- Apply sliding-window rate limiting to protect AI and general API endpoints.
- Apply production security headers.
- Keep Gemini API keys and Supabase service-role credentials off the frontend.
- Do not bypass the server-side Proof Mode AI lockout.

8. Technology and Deployment Environment

Layer	Technology
Frontend	Next.js 16, TypeScript, Tailwind CSS v4, Radix UI / shadcn/ui
Backend	FastAPI 0.115+, Python 3.12/3.14, Pydantic v2
Database & Auth	Supabase PostgreSQL with RLS; Supabase Auth with Google OAuth 2.0 PKCE
AI	Google Gemini 2.5 Flash through server-side provider abstraction
Testing	Pytest and Playwright
Deployment	Docker, Cloudflare Pages, Render / PaaS web services

9. Testing and Verification SOP

- Run the backend unit, integration and security test suite.

- Run frontend linting and production build validation.
- Run Playwright end-to-end browser tests.
- Verify Proof Mode blocks AI requests at the server/API layer.
- Verify ownership and RLS behavior for learning evidence and history.
- Do not open or merge changes until required checks pass.

The project documentation reports 55 passing backend tests and Playwright E2E specifications.

10. Local Execution SOP

- Install Node.js 20+, Python 3.12 or 3.14, and Git. Docker is optional for containerized local execution.
- Clone the repository and enter the PROOFLEARN directory.
- Create and activate the backend Python virtual environment, install dependencies, and configure the .env file.
- Install frontend dependencies and configure .env.local.
- Run backend tests and start the FastAPI development server.
- Start the Next.js frontend and validate the application in the browser.

11. MVP Curriculum Coverage

Subject	Foundational Concepts	Difficulty
Python	Variables & Types; Functions; Lists & Dictionaries	Beginner / Intermediate
Java	Variables & Types; Methods; OOP Basics	Beginner / Intermediate
SQL	SELECT Queries; Table JOINS; Aggregations	Beginner / Intermediate
AI & Machine Learning	Supervised Learning; Model Evaluation; Regularization	Intermediate / Advanced
Data Science	Data Cleaning; Pandas DataFrames; Feature Engineering	Intermediate

12. Development Governance

- Use descriptive branches such as feat/, fix/, docs/, and test/.
- Use Conventional Commit messages.
- Run backend tests, frontend lint/build checks before a pull request.
- Never commit API keys, service-role secrets, passwords, or local environment files.
- Follow the project's Code of Conduct and maintain a respectful, professional environment.

13. SOP Completion / Acceptance Criteria

Checkpoint	Expected Result
Learning	Student receives concept guidance through the learning stage.
Practice	Student completes formative checks before proof.
Proof Mode	Server-side AI access lock is active during proof.
Independent Proof	Student completes the independent challenge without AI.
Transfer	Student demonstrates the concept in a novel context.
Evidence	LEI is generated from the defined evidence dimensions.
History	Verified attempt is stored in the private learning ledger.

Checkpoint	Expected Result
Security	Authentication, RLS, ownership checks, rate limits and secret isolation remain enforced.
Testing	Automated validation passes before release.

14. Source and Documentation Note

This SOP is derived from the uploaded PROOFLEARN README, contributing guidelines, code-of-conduct document, and container configuration. Where those sources did not specify an operational detail, this SOP keeps the wording at the level supported by the project documentation rather than inventing implementation requirements.